



How To Test The Quality Of An Electronic System

An electronic system's quality test engineer is required to work on different electronic products of different technologies and applications throughout his or her career. This article describes the methodology to be followed by an electronics engineer, when he or she is required to test and certify a new system or product, whose technology and application is unknown to him or her. The methodology is given as a series of steps and can be implemented by anybody



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The methodology for testing a new system or product is almost identical to the methodology followed by people while visiting an unknown city/town. Although Google Maps are extensively used nowadays, earlier people relied on paper maps and common sense. They would get around the city/town using a series of famous landmarks, obtained either through enquiry or paper maps. A similar approach can be adopted here, through a series of steps, involving a top-down methodology of going deep into the system. These steps are explained next.

her job is to assure that the manufactured product meets all required specifications.

For example, a transmitter will have a frequency specification as per application requirement, its tolerance, modulation index, spurious and harmonics specific performance requirements, and the like. Typical specifications for a transmitter and a receiver are given in Tables 1 and 2, and the lists are not exhaustive.

3. Power and I/Os. All electronic products require power to perform their respective functions. This power could either be DC or mains AC supply. For example, a TV works on mains AC power and a cellphone works on DC power. Hence, the engineer has to understand the power requirements of the product, whether AC or DC, their values and current requirements from the power source.

Every electronic product will have a number of inputs/outputs (I/Os) for interfacing with other systems or the environment. The I/Os must be understood well. These I/Os could be analogue, digital, radio frequency (RF) or power. The engineer should understand every signal, whether it is input or output.

For analogue signals, dynamic range and bandwidth must be known. He or she should also know whether analogue signals are single-ended or differential.

For digital signals, their type (TTL, ECL, etc), standard RS 232, 422 or 485, and so on must be known. The I/Os are distributed on various types of connectors mounted on the unit. The engineer should know the association between every connector pin and I/Os of the unit. This association is required to arrive at a test setup

Typical Specifications for a Transmitter Table 1

Typical specifications	
Frequency	MHz or GHz
Power O/P	Watts or dBm
Deviation sensitivity	MHz/volt
Spurious and harmonics	dBc
Frequency response or bandwidth	MHz
Supply voltage	Volts DC or AC mains

Typical Specifications for a Receiver Table 2

Typical Specifications	
Frequency	MHz or GHz
Sensitivity	dBm
Dynamic range	dBm
Frequency response or bandwidth	MHz
Supply voltage	Volts DC or AC mains

1. Functionality. No matter how complex an electronics system/product may be, it is designed to perform a specific function or set of functions. The first step in understanding a product is to understand its functionality.

Next comes understanding its application. For example, the product could be a transmitter whose purpose is to modulate information signal on a VHF/UHF/microwave carrier. This product could be used in telecommunications, satellite communications, telemetry measurement systems and so on.

2. Specifications. Specifications define how best a product has to perform its function. These also depend on application requirements. Product specifications must be understood by the quality test engineer, since his or